SAI PRASATH S

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INTERNSHIP

AI Intern, *Ford Motors Private Limited* — Jul 2025 – Present, Chennai, India

- Built **LLM-powered systems** for manufacturing quality analysis, enabling automated issue classification, root cause analysis.
- Developed and deployed **GenAI APIs** using **FastAPI** to integrate AI capabilities into enterprise workflows.
- Optimized **BigQuery data pipelines** operating at GB–TB scale, reducing processing time by approximately **30%**.
- Designed a **RAG-based enterprise knowledge platform** over **2,000+ documents** using hybrid retrieval with **FAISS and BM25**.
- Improved retrieval efficiency, indexing performance, and system reliability through optimized vector indexing and pipeline enhancements.
- Collaborated on AI-driven automation solutions to improve manufacturing quality analysis, knowledge discovery, and decision support.

Project Intern, *Clowak Innovations- VIT-TBI* — Jan 2023 – Mar 2023, Vellore, India

- Developed a **React Native mobile application** for user interaction and real-time system monitoring.
- Integrated **Google Firebase** for authentication, login management, and cloud-based data storage.
- Connected and integrated electronic hardware components including **headlights, side stand, indicators, and sensors** with the instrumentation cluster.
- Collaborated on embedded and mobile integration workflows to improve vehicle instrumentation and monitoring capabilities.

EDUCATION

M.TECH. COMPUTER SCIENCE AND ENGINEERING, — 2024 – present
Vellore Institute of Technology *CGPA: 8.81* — Chennai, India

B.TECH.ELECTRONICS AND INSTRUMENTATION ENGINEERING, — 2020 – 2024
Vellore Institute Of Technology *CGPA : 8.17* — Vellore, India

SKILLS

Languages (Python, Java, SQL, JavaScript, MATLAB) | **Machine Learning/Deep Learning** (Tensorflow, Keras, Scikit-learn, CNN, ANN, NLP) | **GenAI/LLM** (LLMs, RAG, Prompt Engineering, FAISS, pgvector, BM25, Hybrid Retrieval, Langchain) | **Backend** (FastAPI, Flask, Node.js, Express.js) | **Web Development** (HTML, CSS, JavaScript, React) | **Cloud & Databases** (GCP BigQuery, Google Firebase, AWS cloud)

POSITION OF RESPONSIBILITIES

Instrument Society Of India - VIT, *Projects Head* — Jan 2022 – Jan 2023, Vellore, India

- Mentored **20+ students** in technical projects, prototyping, and implementation.
- Led project planning, coordination, and execution for student-led engineering initiatives.
- Organized and supported technical workshops to improve practical learning and hands-on development skills.

PROJECTS

Real-time ECG Analysis and Classification using Neural Networks in IoT devices
- Develop a portable hardware setup for real-time ECG signal acquisition and pre-processing powered by smartphones or laptops. Implement IoT for data transmission to a dedicated computer with a pre-trained neural network for ECG diagnosis, enabling diagnosis results to be sent back to smart devices for easy access.

Assisted Emergency Robot for Fire Brigade, *Arduino, Sensors, Bluetooth, Embedded Systems, Mobile App Integration*
- Developed a **Bluetooth-controlled emergency response robot** equipped with environmental sensors to monitor CO2, nitrogen, temperature, humidity, flame, and smoke levels.
- Built a real-time monitoring interface to display sensor readings and trigger instant alerts for flame and smoke detection.
- Designed the system to assist fire brigade teams in remote inspection of hazardous environments.

Design of FIR Filter using Approximate Computing, *Verilog HDL Implementation*
- Engineered a high-performance FIR filter utilizing approximate computing techniques(HOAANED adder) in Verilog HDL, achieving optimized computational efficiency while maintaining accuracy, thus demonstrating expertise in digital signal processing and hardware description languages.

Bottle Precision Orientation and Adjustment System, *Image processing, Matlab, sensors*
- Designed an **automated bottle orientation and adjustment system** for filling and packaging operations.
- Used **MATLAB-based image processing** to detect bottle position, orientation, and alignment errors.
- Integrated sensors and robotic adjustment mechanisms to ensure consistent and precise bottle positioning.

CNN for Emotion classification in text, *(Python- NLTK,Pandas, Keras, Scikit-learn, Matplotlib)*
- Developed a **CNN-based NLP model** for multi-class emotion classification from text data.
- Built an **end-to-end machine learning pipeline** including text preprocessing, tokenization, stop-word removal, feature extraction, model training, and evaluation.
- Trained and optimized a **CNN architecture using Keras**, tuning hyperparameters to improve classification accuracy.
- Used **Scikit-learn** for model evaluation and performance analysis across multiple emotion classes.
- Improved multi-class prediction performance through preprocessing optimization and model tuning.

Zomato Clone - Frontend, *Html, Css, Javascript*
- Developed a Zomato clone frontend, replicating the user interface and user experience of the Zomato app, enhancing skills in web development, UI design, and creating responsive interfaces.

Mental Health Monitor (MHM) – Multi-Agent LLM System,
(Python, Flask, DistilBERT, LLMs, pgvector, NLP, Vector Embeddings, Chart.js, RBAC)
- Designed and implemented a **multi-agent LLM-driven mental health monitoring system** with context-aware conversational AI for continuous user interactions.
- Engineered a **Long-Context Cognitive Mapping memory layer** using **pgvector and vector embeddings** to retain longitudinal user context, achieving approximately **82% Top-K retrieval accuracy**.
- Built a **DistilBERT + LLM hybrid pipeline** for **4-class sentiment classification**, achieving approximately **88% emotion detection accuracy**.
- Developed a **MIND-SAFE crisis detection framework** to override unsafe LLM responses and trigger real-time human intervention alert workflows.
- Implemented a **real-time analytics dashboard** to monitor user sentiment trends, crisis indicators, conversation insights, and system performance metrics.
- Integrated **role-based access control** to protect sensitive user data, enforce privacy-aware access permissions, and support secure human-in-the-loop intervention.
- Optimized performance through **parallel Flask-based multi-agent orchestration**, reducing latency by approximately **30%** and achieving **sub-2-second response time** with approximately **95% reliability**.

Energy Meter Reading Using Image Processing, *Azure Computer Vision, Google Firebase, HTML, CSS, JavaScript*
- Built a **web-based energy meter reading system** that extracts accurate readings from uploaded meter images using **Azure Computer Vision**.
- Developed a **user-friendly interface** for image upload, reading display, and result verification.
- Integrated **Google Firebase** for secure authentication, database management, and storage of extracted readings.

HACKATHONS

DATASET ' 24 Hackathon

- Participated in a data-centric hackathon involving analytical problem solving, dataset interpretation, and AI/ML-based solution development
- Explored data preprocessing, pattern analysis, and model-oriented approaches for extracting insights from structured and unstructured datasets under competitive timelines
- Strengthened practical experience in teamwork, technical presentation, and rapid experimentation with machine learning workflows

Daimler Hackathon

- Participated in an industry-focused hackathon centered on solving real-world automotive and industrial technology challenges through collaborative problem-solving and rapid prototyping
- Worked in a cross-functional team environment to analyze use cases, develop technical approaches, and present scalable solution ideas within constrained timelines
- Gained exposure to enterprise manufacturing workflows, industrial systems, and large-scale operational processes through technical sessions and plant interaction activities

CERTIFICATES

Full Stack Web Development - SHAPEAI

Python - GUVI

AWS Solution Architect Associate

PUBLICATIONS

Real-time ECG Analysis and Classification using Neural Networks in IoT devices,

IEEE i-PACT 2023 | DOI: 10.1109/i-PACT58649.2023.10434526

- Developed a **portable hardware setup** for real-time ECG signal acquisition and preprocessing, powered by smartphones or laptops.
- Implemented an **IoT-based data transmission pipeline** to send ECG signals to a dedicated system for analysis.
- Used a **pre-trained neural network model** for ECG classification and diagnostic prediction.
- Enabled diagnosis results to be transmitted back to smart devices for accessible and real-time health monitoring.